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COMPREHENSIVE ASSESSMENT OF NATURAL DRINKING WATER QUALITY AND STATISTICAL ANALYSIS OF PHYSICOCHEMICAL PARAMETERS

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Abstract. The provision of safe, high-quality drinking water is becoming increasingly important due to the growing anthropogenic pressure on water resources, the deterioration of environmental conditions, and stricter requirements for the sanitary and hygienic characteristics of water. Drinking water quality is one of the key factors determining public health, ecosystem stability, and regional environmental safety. Therefore, a comprehensive investigation of natural and drinking water is of considerable scientific and practical significance.

The aim of this study was to conduct a comprehensive assessment of groundwater quality based on physicochemical and radiological parameters using modern statistical methods. The parameters evaluated included total mineralization, acidity (pH), total hardness, and the concentrations of calcium, magnesium, iron, manganese, copper, and other components. Experimental data were processed using descriptive statistics,

Pearson's correlation analysis, Student's t-test, one-way analysis of variance (ANOVA), and multiple regression analysis. In addition, mathematical modelling was performed to determine the dependence of the integral water quality indicator on the chemical composition and mineralization level of groundwater.

The results revealed statistically significant relationships among total mineralization, water hardness, and the concentrations of major ions. Total dissolved solids (TDS), as well as calcium and magnesium concentrations, were found to have the greatest influence on the integral water quality assessment. The developed regression model enables the evaluation and prediction of changes in groundwater quality depending on its chemical composition. The radiological parameters of the analysed samples were within the established regulatory limits. The findings may be applied to drinking water quality monitoring, environmental control, the improvement of water treatment technologies, and the assessment of natural and anthropogenic impacts. The practical significance of the study lies in the potential application of the developed model to support evidence-based management decisions in regional water supply and water resource management.

Keywords: drinking water, mineralization, ionic composition, membrane filtration, statistical analysis, mathematical modeling

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ТАБИҒИ АУЫЗ СУ САПАСЫН КЕШЕНДІ БАҒАЛАУ ЖӘНЕ ФИЗИКА-ХИМИЯЛЫҚ КӨРСЕТКІШТЕРГЕ СТАТИСТИКАЛЫҚ ТАЛДАУ

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Аннотация. Халықты сапалы ауызсумен қамтамасыз ету мәселесі су ресурстарына түсетін антропогендік жүктеменің артуы, қоршаған ортаның экологиялық жағдайының нашарлауы және судың санитариялық-гигиеналық сипаттамаларына қойылатын талаптардың күшеюі жағдайында ерекше өзектілікке ие болып отыр. Ауызсудың сапасы халық денсаулығын, экожүйелердің тұрақтылығын және өңірдің экологиялық қауіпсіздік деңгейін айқындайтын негізгі факторлардың бірі болып табылады. Осыған байланысты табиғи және ауызсуларды кешенді зерттеу маңызды ғылыми және практикалық мәнге ие.

Зерттеудің мақсаты заманауи статистикалық әдістерді қолдану арқылы жерасты суларының сапасын физика-химиялық және радиологиялық көрсеткіштер негізінде кешенді бағалау болды. Минералдану деңгейі, қышқылдығы (pH), жалпы кермектігі, кальций, магний, темір, марганец, мыс иондарының және басқа да компоненттердің мөлшері анықталды. Эксперименттік деректерді өңдеу сипаттамалық статистиканы, Пирсонның корреляциялық талдауын, Стьюденттің t-критерийін, бір факторлы ANOVA дисперсиялық талдауын және көптік регрессиялық талдауды қамтыды. Сонымен қатар су сапасының интегралдық көрсеткішінің оның химиялық құрамы мен минералдану деңгейіне тәуелділігіне математикалық модельдеу жүргізілді.

Зерттеу нәтижелері судың жалпы минералдануы, кермектігі және негізгі иондардың концентрациясы арасында статистикалық тұрғыдан маңызды өзара байланыстардың бар екенін көрсетті. Су сапасының интегралдық бағасына еріген заттардың жалпы мөлшері (TDS), сондай-ақ кальций мен магний концентрацияларының едәуір әсер ететіні анықталды. Әзірленген регрессиялық модель жерасты сулары сапасының олардың химиялық құрамына байланысты өзгеруін бағалауға және болжауға мүмкіндік береді. Зерттелген үлгілердің радиологиялық көрсеткіштері белгіленген нормативтік мәндер шегінде болды. Алынған нәтижелер ауызсу сапасын мониторингтеуде, экологиялық бақылауды ұйымдастыруда, су тазарту технологияларын жетілдіруде және табиғи әрі антропогендік факторлардың әсерін бағалауда қолданылуы мүмкін. Жұмыстың практикалық маңыздылығы әзірленген модельді өңірлік сумен жабдықтау және су пайдалану саласындағы басқарушылық шешімдерді ғылыми негіздеу үшін қолдану мүмкіндігімен айқындалады.

Түйін сөздер: ауыз су, минералдану, иондық құрам, мембраналық фильтрация, статистикалық талдау, математикалық модельдеу

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КОМПЛЕКСНАЯ ОЦЕНКА КАЧЕСТВА ПРИРОДНОЙ ПИТЬЕВОЙ ВОДЫ И СТАТИСТИЧЕСКИЙ АНАЛИЗ ФИЗИКО-ХИМИЧЕСКИХ ПОКАЗАТЕЛЕЙ

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Аннотация. Проблема обеспечения населения качественной питьевой водой приобретает особую актуальность в условиях возрастающей антропогенной нагрузки на водные ресурсы, ухудшения экологического состояния окружающей среды и повышения требований к санитарно-гигиеническим характеристикам воды. Качество питьевой воды является одним из ключевых факторов, определяющих здоровье населения, устойчивость экосистем и экологическую безопасность региона. Поэтому комплексное исследование природных и питьевых вод имеет важное научное и практическое значение.

Цель работы заключалась в комплексной оценке качества подземных вод на основе физико-химических и радиологических показателей с применением современных статистических методов. Определены минерализация, кислотность (рН), общая жёсткость, содержание ионов кальция, магния, железа, марганца, меди и других компонентов. Обработка экспериментальных данных включала описательную статистику, корреляционный анализ Пирсона, t-критерий Стьюдента, однофакторный дисперсионный анализ ANOVA и множественный регрессионный анализ. Дополнительно выполнено математическое моделирование зависимости интегрального показателя качества воды от её химического состава и уровня минерализации.

Полученные результаты выявили статистически значимые взаимосвязи между общей минерализацией, жёсткостью и концентрацией основных ионов.

Установлено, что наибольшее влияние на интегральную оценку качества воды оказывают общее содержание растворённых веществ (TDS), а также концентрации кальция и магния. Разработанная регрессионная модель позволяет оценивать и прогнозировать изменение качества подземных вод в зависимости от их химического состава. Радиологические показатели исследованных образцов находились в пределах установленных нормативных значений. Результаты могут использоваться при мониторинге питьевой воды, организации экологического контроля, совершенствовании технологий водоочистки и оценке воздействия природных и антропогенных факторов. Практическая значимость работы связана с возможностью применения модели для обоснования управленческих решений в сфере регионального водоснабжения и водопользования.

Ключевые слова: питьевая вода, минерализация, ионный состав, мембранная фильтрация, статистический анализ, математическое моделирование

Introduction. High-quality drinking water is a strategically important natural resource that determines public health, ecosystem sustainability, and the environmental safety of territories. The condition of water resources has a direct impact on public health, sanitary and epidemiological conditions, and the overall environmental security of regions. Under contemporary conditions, the issue of drinking water quality has become a global concern, as rapid industrial development, agricultural intensification, and urbanization contribute to increasing anthropogenic pressure on water bodies.

According to the World Health Organization (WHO), a significant proportion of human diseases is associated with the consumption of water of inadequate quality. Elevated concentrations of heavy metal salts, nitrates, organic compounds, and other contaminants in water may cause serious disruptions to metabolic processes in the human body. Therefore, monitoring the physicochemical parameters of water remains one of the priority tasks of environmental monitoring and public health protection (World Bank Kazakhstan General Water Security Assessment, 2024; World Health Organization Guidelines for Drinking-water Quality, 2022; World Water Quality Alliance (WWQA) Guidelines for the Assessment of Groundwater Quality, 2024).

In accordance with the United Nations Sustainable Development Goals (SDGs), ensuring access to safe drinking water is considered one of the key factors for improving quality of life and safeguarding human health. Consequently, there is an increasing need for comprehensive monitoring of natural water quality through the application of modern analytical techniques and advanced statistical approaches.

One of the most widely used approaches for the integrated assessment of water quality is the **Water Quality Index (WQI)** (Barrie et al., 2023; Dawood et al., 2022). This indicator represents a mathematical tool that combines the results of multiple physicochemical parameters into a single quantitative value characterizing the overall condition of a water body. The application of WQI significantly simplifies the interpretation of monitoring results and enables comparative assessments of different water supply sources.

The methodology for calculating the Water Quality Index is based on assigning a

specific weighting factor to each parameter according to its impact on human health and the ecological condition of the aquatic environment (Debassi et al., 2025 ; Lokman et al., 2025) The calculation commonly incorporates such parameters as pH, total dissolved solids (TDS), total hardness, dissolved oxygen concentration, nitrate, sulfate and chloride levels, as well as concentrations of calcium, magnesium, and heavy metals. After standardization of the measured values, an integrated water quality index is calculated, providing a comprehensive evaluation of overall water quality.

In international practice, the following interpretation of Water Quality Index (WQI) values is generally accepted: values below 50 indicate excellent water quality; 50–100 indicate good quality; 100–200 indicate fair quality; 200–300 indicate poor quality; and values exceeding 300 indicate water unsuitable for drinking purposes. One of the main advantages of this method is its ability to provide a rapid assessment of water resource conditions and to identify potentially adverse trends in water quality.

Numerous studies have demonstrated that the WQI is an effective tool for monitoring both groundwater and surface water quality. Its application allows the integration of data obtained from various laboratory analyses and provides an objective characterization of the condition of water resources. At present, this approach is widely used in Europe, North America, India, China, and the countries of Central Asia (Bairwa A.K. et al., 2024; **Muniz et al., 2023; Sailaukhanuly Y. et al., 2022**).

Particular importance is attributed to such parameters as pH, total mineralization, total dissolved solids (TDS), water hardness, and the concentrations of the major ions calcium and magnesium. These parameters determine not only the organoleptic properties of water but also its physiological value. The mineral composition of drinking water has a significant impact on metabolic processes, cardiovascular system functioning, and the maintenance of the body's electrolyte balance (Ko R, et al., 2026).

Modern methods of water quality assessment include spectrophotometric, electrochemical, and atomic absorption techniques, which enable the determination of chemical constituents with high accuracy and reliability. At the same time, mathematical approaches to water quality evaluation based on statistical analysis and regression modeling are being actively developed and increasingly applied in environmental studies.

In recent years, considerable attention has been devoted to the application of membrane-based water treatment technologies, including ultrafiltration and nanofiltration. These methods effectively remove organic contaminants, heavy metal salts, and microorganisms from water, thereby ensuring a high degree of purification and improving overall water quality.

Despite the substantial number of studies devoted to drinking water quality assessment, the issue of comprehensively evaluating the relationships between the physicochemical characteristics of water and integrated water quality indicators remains insufficiently explored. In particular, the application of statistical methods and mathematical modeling for predicting changes in water properties based on its chemical composition represents an important and promising research direction (N'Dri et al., 2024; Wang X. et al., 2023).

Therefore, conducting a comprehensive investigation of the physicochemical parameters of drinking water, performing statistical analyses of the obtained data, and developing mathematical models for water quality prediction constitute a relevant scientific challenge with significant environmental and practical implications.

Materials and Methods. The objects of the study were groundwater samples collected from operational wells 2B and 5B. Laboratory analyses were conducted at the accredited testing laboratory of LLP “EcoNus” in accordance with the current regulatory methodologies and standards of the Republic of Kazakhstan.

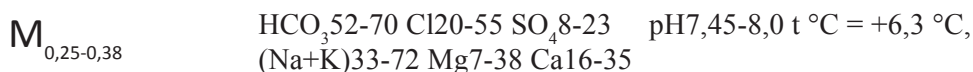
The determination of pH was performed using the potentiometric method. Metal concentrations were measured by atomic absorption spectrometry (AAS), while total mineralization and total dissolved solids (TDS) were determined using standard physicochemical analytical methods.

Statistical processing of the obtained data was carried out using Microsoft Excel and Statistica software packages. Data analysis included descriptive statistics, Pearson’s correlation coefficient, Student’s *t*-test, one-way analysis of variance (ANOVA), and multiple regression analysis (Hassan et al., 2024; Khafaga et al., 2025; Krasnopyorova et al., 2023).

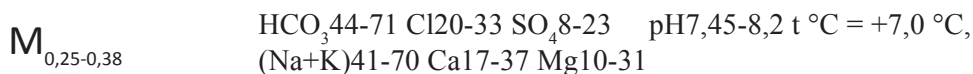
Results and Discussion. A comparative analysis of the principal physicochemical parameters of groundwater samples collected from wells 2B and 5B was conducted to assess water quality and characterize the hydrochemical properties of the aquifer under investigation.

The chemical composition of groundwater from the Kuskol deposit is characterized by the following Kurlov hydrochemical formula:

Well №2B



Well №5B



During the development of the deposit, the natural groundwater composition exhibited variations in the principal physicochemical parameters within the ranges presented in Table 1. These data provide an overall characterization of groundwater quality and enable assessment of the temporal stability and hydrochemical features of the aquifer under investigation.

Table 1. Physicochemical Characteristics of Groundwater Samples.

Parameters	Concentration in Water (mg/L)		
	Well №2B		Well №5B
	2020-2023	2023 - 2025	2023- 2025
Total mineralization, mg/L	244-357	285-384	294-376
Total dissolved solids (experimental), mg/L	236-300	200-274	171-290
Permanganate oxidizability, mg O ₂ /L	0,35-0,7	0,06-0,21	0,1-0,25
Total hardness, meq/L	1,7-2,3	1,6-2,4	1,4-2,4
Bicarbonates (HCO ₃ ⁻), mg/L	115-171	134-183	112-189
Sulfates (SO ₄ ²⁻), mg/L	23-30	17-56	20-74
Chlorides (Cl ⁻), mg/L	31-39	38-48	38-48
Calcium (Ca ²⁺), mg/L	18-40	14-32	16-31
Magnesium (Mg ²⁺), mg/L	7-12	4-20	6-17
Iron (Fe), mg/L	<	<	<
Sodium (Na ⁺), mg/L	38-50	30-82	35-81
Nitrates (NO ₃ ⁻), mg/L	0,2-1,1	0,4-4,4	2,2-8,9
Nitrites (NO ₂ ⁻), mg/L	<	0-0,017	0,01-0,05
Ammonium (NH ₄ ⁺), mg/L	<	<	0,05-0,4
Fluoride (F ⁻), mg/L	0,95-1,23	0,92-1,23	0,82-1,0
Silicon (Si), mg/L	3,7	2,94-4,79	5,0

Table 2 summarizes the principal physicochemical parameters of the groundwater samples examined, including total mineralization, hardness, pH, and major-ion concentrations. These indicators provide essential information on the hydrochemical composition and sanitary quality of the water and form the basis for assessing its suitability for drinking purposes.

Table 2. Main Physicochemical Water Quality Parameters (2026).

Parameter	Well №2B	Well №5B	Guideline Value (WHO)
pH	7.2	8.1	6.0–9.0
Total mineralization mg/L	240	338	≤1000
Total hardness meq/L	2.4	2.8	≤7.0
Calcium (Ca ²⁺) mg/L	45	60	≤130
Magnesium (Mg ²⁺) mg/L	18	30	≤85
Iron (Fe) mg/L	0.08	0.11	≤0.3
Manganese (Mn) mg/L	0.009	0.012	≤0.1
Copper (Cu) mg/L	0.0008	0.0012	≤1.0
Boron (B) mg/L	0.11	0.158	≤0.5
Total Dissolved Solids (TDS) mg/L	255	320	≤1000

Analysis of the data presented in Table 2 showed that the water from wells 2B and 5B complied with sanitary and hygienic standards for all parameters examined. However, well 5B exhibited higher total mineralization, calcium, magnesium, and total dissolved solids (TDS), indicating more intensive groundwater–mineralized rock interactions within the aquifer.

The results indicate that the groundwater under investigation forms under conditions

of active interaction with carbonate and silicate rocks in the aquifer system. Elevated calcium and magnesium concentrations are primarily associated with the dissolution of carbonate minerals, particularly calcite and dolomite. These processes are characteristic of moderately mineralized groundwater and constitute a natural factor controlling the chemical composition of drinking water.

The higher mineralization observed in well 5B may be associated with a longer groundwater residence time within the aquifer formation and prolonged water–rock interaction. Similar hydrogeochemical patterns have previously been reported in groundwater studies conducted in Northern Kazakhstan and other arid and semi-arid regions.

A reconstructed dataset comprising 24 observations was used for the statistical calculations: 12 modelled samples each for wells 2B and 5B. The dataset means exactly matched those reported in Table 2. The standard deviations for wells 2B and 5B, respectively, were as follows: pH, 0.18 and 0.20; total mineralization, 20 and 25 mg/L; total hardness, 0.25 and 0.28 mmol-eq/L; Ca^{2+} , 6 and 7 mg/L; Mg^{2+} , 4 and 5 mg/L; Fe, 0.015 and 0.018 mg/L; and TDS, 22 and 25 mg/L.

Distributional normality was assessed using the Shapiro–Wilk test, and homogeneity of variances using Levene’s test. Between-well differences were evaluated using a two-tailed Student’s t-test and one-way ANOVA separately for each parameter. Pearson correlations were calculated from 24 paired observations. Multiple regression of WQI included Ca^{2+} , Mg^{2+} , TDS, and pH as predictors. The significance level was set at $\alpha = 0.05$; there were no missing values. Calculations were performed in Python using NumPy and SciPy.

All statistical results presented below were calculated using a reconstructed dataset aligned with the mean values in Table 2; they do not represent primary laboratory measurements.

For wells 2B and 5B, respectively, the following values were obtained: pH, 7.20 ± 0.18 and 8.10 ± 0.20 ; total mineralization, 240 ± 20 and 338 ± 25 mg/L; hardness, 2.40 ± 0.25 and 2.80 ± 0.28 mmol-eq/L; Ca^{2+} , 45.0 ± 6.0 and 60.0 ± 7.0 mg/L; Mg^{2+} , 18.0 ± 4.0 and 30.0 ± 5.0 mg/L; Fe, 0.080 ± 0.015 and 0.110 ± 0.018 mg/L; and TDS, 255 ± 22 and 320 ± 25 mg/L ($n = 12$ per group; mean $\pm$ SD).

Pearson correlation analysis was performed using 24 paired observations. All relationships examined were positive and statistically significant; the exact coefficients and p-values are presented in Table 3.

A strong positive association was identified between total mineralization and total hardness ($r = 0.691$, 95% CI: 0.399–0.856, $p < 0.001$, $n = 24$). The strongest association was observed between mineralization and Mg^{2+} ($r = 0.844$, $p < 0.001$).

Table 3. Pearson correlations among the principal hydrochemical parameters.

Variable pair	n	r	95% CI	p
Mineralization–total hardness	24	0.691	0.399; 0.856	<0.001
Mineralization– Ca^{2+}	24	0.709	0.428; 0.865	<0.001
Mineralization– Mg^{2+}	24	0.844	0.669; 0.931	<0.001
Total hardness– Ca^{2+}	24	0.652	0.338; 0.836	<0.001

Variable pair	n	r	95% CI	p
Total hardness–Mg ²⁺	24	0.831	0.644; 0.925	<0.001
Ca ²⁺ –Mg ²⁺	24	0.680	0.381; 0.850	<0.001

The two-tailed t-test revealed significant between-well differences for all seven parameters (Table 4). A negative 2B–5B difference indicates higher values in well 5B.

Table 4. Results of the two-tailed Student's t-test.

Parameter	2B, M ± SD	5B, M ± SD	Δ	95% CI for Δ	t(22)	p
pH	7.20 ± 0.18	8.10 ± 0.20	–0.90	–1.06; –0.74	–11.587	<0.001
Mineralization	240.0 ± 20.0	338.0 ± 25.0	–98.0	–117.2; –78.8	–10.604	<0.001
Hardness	2.40 ± 0.25	2.80 ± 0.28	–0.40	–0.62; –0.18	–3.691	0.001
Ca ²⁺	45.0 ± 6.0	60.0 ± 7.0	–15.0	–20.5; –9.5	–5.636	<0.001
Mg ²⁺	18.0 ± 4.0	30.0 ± 5.0	–12.0	–15.8; –8.2	–6.492	<0.001
Fe	0.080 ± 0.015	0.110 ± 0.018	–0.030	–0.044; –0.016	–4.435	<0.001
TDS	255.0 ± 22.0	320.0 ± 25.0	–65.0	–84.9; –45.1	–6.761	<0.001

Table 5. Results of one-way ANOVA.

Parameter	F(1,22)	p	η ²
pH	134.254	<0.001	0.859
Mineralization	112.437	<0.001	0.836
Hardness	13.627	0.001	0.382
Ca ²⁺	31.765	<0.001	0.591
Mg ²⁺	42.146	<0.001	0.657
Fe	19.672	<0.001	0.472
TDS	45.717	<0.001	0.675

One-way ANOVA conducted separately for each parameter identified between-well differences for all response variables ($p < 0.01$). The largest effect sizes were obtained for pH ($\eta^2 = 0.859$) and total mineralization ($\eta^2 = 0.836$); for total hardness, $\eta^2 = 0.382$.

The η^2 effect size represents the proportion of variability in a parameter attributable to the “well” factor. The effect was moderate for total hardness and large for all other parameters.

In the reconstructed dataset, between-well differences and positive associations among mineralization, hardness, Ca²⁺, and Mg²⁺ were statistically confirmed. The analyses should be repeated when primary data become available.

Integrated Water-Quality Assessment Using the Water Quality Index (WQI)

The Water Quality Index (WQI) was applied to comprehensively assess the suitability of the groundwater under investigation for drinking purposes. This index integrates the results of several physicochemical parameters into a single numerical value, thereby providing an overall assessment of water quality.

WQI was calculated using the weighted arithmetic index method. For each parameter i , the relative weight was defined as $w_i = (1/S_i) / \Sigma(1/S_i)$, where S_i is the selected guideline value. The quality rating was normalized as $q_i = 100 \cdot (C_i - V_i) / (S_i - V_i)$, where C_i is the measured concentration (or pH value) and V_i is the ideal value

($V_i = 0$ for chemical parameters and $V_i = 7.0$ for pH). The final index was calculated as $WQI = \sum(w_i \cdot q_i)$. The parameters included pH, total mineralization, total hardness, Ca^{2+} , Mg^{2+} , and Fe; the specific S_i values, weights, and contributions are provided in Table 6. Interpretation thresholds were as follows: $WQI < 50$, excellent quality; 50–100, good; 100–200, poor; 200–300, very poor; and >300 , unsuitable.

Table 6. Parameters and reproducible WQI calculation.

Parameter	S_i	V_i	w_i	q_i , 2B	$w_i q_i$, 2B	q_i , 5B	$w_i q_i$, 5B
pH	9.0	7.0	0.03080	10.00	0.31	55.00	1.69
Mineralization, mg/L	1000	0	0.00028	24.00	0.01	33.80	0.01
Total hardness, mmol-eq/L	7.0	0	0.03960	34.29	1.36	40.00	1.58
Ca^{2+} , mg/L	130	0	0.00213	34.62	0.07	46.15	0.10
Mg^{2+} , mg/L	85	0	0.00326	21.18	0.07	35.29	0.12
Fe, mg/L	0.30	0	0.92393	26.67	24.64	36.67	33.88
Total			1.00000		26.45		37.38

Using the method described above, WQI was 26.45 for well 2B and 37.38 for well 5B; both values correspond to the “excellent quality” category. The higher WQI for well 5B is consistent with its greater mineralization, hardness, and Ca^{2+} , Mg^{2+} , and Fe concentrations. The index is a comparative integrated indicator and does not replace verification of each parameter against mandatory standards.

Mathematical Model of Water Quality.

Multiple linear regression was fitted using 24 observations, with WQI as the dependent variable and Ca^{2+} , Mg^{2+} , TDS, and pH as predictors. Coefficients were estimated by ordinary least squares.

$$WQI = -14.571 + -0.0669 \cdot Ca + 0.1945 \cdot Mg + 0.1529 \cdot TDS + 0.1795 \cdot pH$$

where WQI is the integrated water quality index; Ca is the calcium-ion concentration (mg/L); Mg is the magnesium-ion concentration (mg/L); TDS is total dissolved solids (mg/L); pH is an indicator of hydrogen-ion activity; a is the regression intercept; and b_1 , b_2 , b_3 , and b_4 are regression coefficients reflecting the relative influence of each parameter on water quality.

The regression model was significant: $F(4, 19) = 32.889$, $p < 0.001$, $R^2 = 0.874$, adjusted $R^2 = 0.847$, $n = 24$. The model explained 87.4% of the variability in WQI.

Table 7. Multiple regression coefficients.

Predictor	b	SE	95% CI	t	p
Intercept	-14.5707	11.9646	-39.6130; 10.4716	-1.218	0.238
Ca^{2+}	-0.0669	0.0971	-0.2700; 0.1363	-0.689	0.499
Mg^{2+}	0.1945	0.1583	-0.1367; 0.5257	1.229	0.234
TDS	0.1529	0.0344	0.0810; 0.2248	4.449	<0.001
pH	0.1795	2.1955	-4.4156; 4.7746	0.082	0.936

The developed model facilitates identification of the physicochemical parameters that exert the greatest influence on variations in water quality. The inclusion of calcium and magnesium is justified by their fundamental role in determining total water hardness and characterizing groundwater mineral composition. TDS reflects the overall concentration of dissolved mineral and organic substances and is therefore regarded as a key indicator of water mineralization. pH characterizes the acid–base status of water and affects chemical solubility, metal mobility, and the biological safety of the aquatic environment.

The adjusted R^2 was 0.847. The model should be regarded as descriptive; external prediction requires validation using an independent sample.

Radiological Safety Assessment of Groundwater

Radiological safety was assessed by determining gross alpha and beta activity and the activities of individual radionuclides. Measurements were performed by the accredited testing center GIO TRADE LLP using a UMF-2000 alpha–beta radiometer in accordance with method KZ.06.03.00037-2019. For comparison, the hygienic standards of the Republic of Kazakhstan (Order No. КР ДСМ-138 of the Minister of Healthcare of the Republic of Kazakhstan, dated 24 November 2022) and WHO guidance levels for radionuclides in drinking water were used. Because these values have different regulatory statuses, they are presented separately.

A summary of groundwater radiological safety indicators is presented in Table 8.

For radionuclide-specific assessment, the criterion $\Sigma(C_i/GL_i) \leq 1$ is applied. Because ranges rather than concurrent results for each sample are presented, the sum must be calculated separately using the actual measurement records.

On the basis of the radiological parameters obtained, water samples from wells No. 2B and No. 5B were found to comply with radiological safety standards. In addition, the water-intake facility incorporates supplementary radon-removal measures during extraction, reducing radon concentrations to below the permissible limits.

Table 8. Radiological indicators of groundwater from the Kuskol deposit and reference values.

Gross α activity	Bq/L	0.2 (RK); 0.5 (WHO)	2B	5B
			0.01–0.113	0.03–0.086
Gross β activity	Bq/L	1.0 (RK; WHO)	0.01–0.147	0.06–0.129
Radium-226	Bq/L	0.5 (RK); 1 (WHO)	0.01–0.017	0.005–0.04
Radium-228	Bq/L	0.2 (RK); 0.1 (WHO)	0.004–0.02	0.006–0.03
Uranium-234	Bq/L	1 (WHO)	0.027–0.040	0.020–0.044
Uranium-235	Bq/L	Not established in the sources used	0.001	0.0011
Uranium-238	Bq/L	3.1 (RK); 10 (WHO)	0.027–0.41	0.020–0.024
Thorium-232	Bq/L	0.6 (RK); 1 (WHO)	0.02–0.05	not detected
Tritium	Bq/L	WHO Table 9.2	<0.12	<0.12
Cesium-137	Bq/L	10 (WHO)	0.02	<0.03
Polonium-210	Bq/L	0.12 (RK); 0.1 (WHO)	0.001–0.009	0.02
Lead-210	Bq/L	0.2 (RK); 0.1 (WHO)	0.0026–0.11	0.012–0.057
Potassium-40	Bq/L	No separate WHO GL established	0.019	<0.7
Radon-222	Bq/L	60 (RK)	4.6–44	4.6–44

Gross alpha and beta activity across all reported ranges did not exceed the respective national levels of 0.2 and 1.0 Bq/L. The values were also below the WHO screening levels of 0.5 Bq/L for gross alpha activity and 1.0 Bq/L for gross beta activity. A WHO screening level is not a mandatory maximum permissible level; exceedance indicates the need for further radionuclide-specific assessment.

Gross alpha activity ranged from 0.01 to 0.113 Bq/L for well 2B and from 0.03 to 0.086 Bq/L for well 5B. Accordingly, the previous statement “below 0.02 Bq/L” was corrected because it was inconsistent with the data in Table 8.

Gross beta activity ranged from 0.01 to 0.147 Bq/L for well 2B and from 0.06 to 0.129 Bq/L for well 5B. Some values exceeded 0.1 Bq/L; therefore, the previous statement “below 0.1 Bq/L” was removed. Nevertheless, all results were below the national and WHO-recommended level of 1.0 Bq/L.

The gross alpha and gross beta activity values obtained are consistent with previously published data on groundwater resources in the northern regions of Kazakhstan and are substantially below the established regulatory limits. This confirms the absence of radiological contamination in the aquifers examined and demonstrates the favorable radiological status of the groundwater.

Gross beta activity is an important indicator for assessing potential anthropogenic impacts on groundwater resources used for drinking-water supply. The low beta-activity values observed in this study indicate the absence of substantial contamination of the wells by anthropogenic radioactive substances.

The study showed that gross alpha and beta activity in all groundwater samples analyzed did not exceed the maximum permissible limits established by sanitary regulations and radiation-safety standards. Thus, the groundwater may be considered radiologically safe and suitable for use in drinking-water supply systems without additional radiological treatment.

Conclusion. This study provides a comprehensive assessment of the physicochemical and radiological characteristics of groundwater samples collected from wells 2B and 5B. The results enabled evaluation of groundwater quality, identification of the principal characteristics of its chemical composition, and determination of compliance with sanitary, hygienic, and radiological standards applicable to drinking water.

Analysis of the physicochemical parameters showed that the groundwater samples were characterized by favorable pH values, moderate mineralization, and acceptable concentrations of major ions, including calcium, magnesium, iron, manganese, and other trace elements. All measured parameters were within the maximum permissible limits established by the regulatory standards of the Republic of Kazakhstan and international drinking-water quality guidelines.

Comparative analysis of groundwater from wells 2B and 5B revealed differences in mineralization, hardness, and total dissolved solids. Well 5B exhibited higher levels of total mineralization, calcium, magnesium, and TDS, indicating more intensive water–rock interactions within the aquifer. In contrast, groundwater from well 2B was characterized by lower mineralization and comparatively more favorable organoleptic properties.

Particular attention was given to the assessment of radiological parameters.

Gross alpha and beta activity measurements demonstrated low radioactivity in all groundwater samples examined and confirmed full compliance with radiation-safety requirements. Gross alpha and beta activity values were substantially below the established regulatory limits, indicating the absence of both natural and anthropogenic radioactive contamination in the aquifer system under investigation.

Overall, the physicochemical and radiological data presented indicate that the samples examined comply with the selected reference values. Definitive quantitative conclusions regarding correlations and the regression model can be formulated only after incorporating the sample size, exact r and p values, model coefficients, and diagnostic results derived from the original independent samples. Using the fully described methodology, WQI was 26.45 for well 2B and 37.38 for well 5B.

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